

NAPLAN

2018 State report – Year 9

February 2019

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Preface

State reports are issued by the QCAA about the performance of Queensland students on the National Assessment Program — Literacy and Numeracy (NAPLAN) paper tests. State reports provide system-level information and are publicly available. This report for Year 9 students in 2018 contains:

- the Queensland performance on each item
- the national performance on each item
- the item descriptors
- a commentary on the state results
- some recommendations for teaching.

Who should use this State report?

The NAPLAN State reports help principals, teachers and other school personnel understand, interpret and use information about student performance on NAPLAN.

School principals can use this report to provide information to the school community on aspects of the tests. This would allow professional conversations with their teachers, curriculum leaders, and department heads. Curriculum leaders can use this information to interpret the more specific information given in their school and class reports. These other reports are explained below.

Since this report is publicly available on the QCAA website, it can also inform providers of teacher training, special education services and educational research and policy.

Parents and carers can use this report to interpret the results on their child's student report. They are also able to judge how their child performed when compared with the whole population of students. The item descriptors provide them with useful information about the scope of the tests.

About the tests

The purpose of the National Assessment Program (NAP) is to collect information that governments, education authorities and schools can use to identify indicative literacy and numeracy skills Australian students can demonstrate. As part of that program, the NAPLAN tests are administered to full cohorts of students in Years 3, 5, 7 and 9 each year. These standardised tests are sources of information about student learning that can be used to inform educational policy and current educational practice.

The NAPLAN tests were initially developed using the nationally agreed Statements of Learning for English and Statements of Learning for Mathematics, 2005. Since 2016 however, the tests are referenced to the Australian Curriculum. The NAPLAN tests are designed to assess student understanding in the following areas:

- Language conventions: The test assesses the ability of students to independently recognise and use correct Standard Australian English grammar, punctuation and spelling.
- Writing: The test assesses the ability of students to convey thoughts, ideas and information through the independent construction of a written text in Standard Australian English.
- Reading: The test assesses the ability of students to independently make meaning from written Standard Australian English texts, including those with some visual elements.

- **Numeracy:** The test assesses students' knowledge of mathematics, their ability to apply that knowledge in context independently, and their ability to reason mathematically.

Marking and scoring the tests

Marking the tests

Markers mark those test items that do not use a multiple-choice format. These markers apply nationally agreed marking guides. There are marking guides for open-ended Reading items if any such items are included. Marking guides allow consistent and reliable judgments by markers. There are guides for the Writing test and one each for the constructed responses in Numeracy and Spelling. For some Numeracy items, students may provide a correct response in different forms. Professional officers decide on agreed scoring protocols for these items.

Calculating raw scores

The simplest calculation made in scoring the tests is the raw score — the number of questions answered correctly. Each of the questions for the Language conventions, Reading and Numeracy tests are marked as either correct or incorrect. Raw scores for the Writing test are sums of the marks on each of ten criteria.

Raw scores have limited use. They enable the performance of students who have all completed the same test at the same time to be placed in a rank order, but they do not provide information about the level of difficulty of the test nor the relative differences between students.

Constructing scaled scores and bands

To make raw scores more useful, they are transferred to scores on a common scale that reflects how difficult it was to achieve each score. Each year ACARA publishes equivalence tables that allow a student's raw score to be located on the NAPLAN scale. The scale is comparable between year levels for each assessment area. An equating process is also carried out on each year's test to enable scores to be compared between successive years of testing. For example, a raw score of 20 on the Year 3 Reading test might be transformed to a scaled score of 354. This will also represent the same achievement for a student with the same scaled score in Year 5, and for a student with the same scaled score for Reading in a previous year.

Each NAPLAN scale is divided into ten bands used to report student progress.

Using NAPLAN reports to inform teaching and learning

Using scaled scores

The scaled score can compare the results of different students. Scaled scores provide a basis for measuring and comparing students' abilities across years of schooling, for example comparing a student's result in Year 3 in 2016 and Year 5 in 2018. The scales can thus help to monitor the growth of groups of students over time. This enables the school to review and/or consolidate special programs that may have been put in place.

Principals and teachers should take care when making comparisons between small groups of students. For groups of fewer than ten students, differences may not be reliable, particularly small differences.

Using item analysis

While the national and state reports provide the comparative data, class reports provide a school with the information that can be used to inform teaching and learning and to build capacity in schools. Analysis of the NAPLAN class data, in particular the performance on each item, will

provide teachers with information about the understandings and patterns of misunderstandings of students.

Looking at the performance on the items and then analysing the error patterns allows teachers and principals to make hypotheses about why groups of students make particular errors. As mentioned below, more detailed analysis by QCAA staff is available from the QCAA website.

Steps for analysis might be as follows:

- Compare the facility rates (percentage correct) achieved by the school's students with the national and state results available in this document. Is their performance consistent?
- Look at the common errors made by their students and compare them with the common errors made in the state. Only errors from Queensland students are available and are found in the item analyses that are part of SunLANDA Online.
- Form hypotheses about why students are making these errors, e.g.
 - How did students think about this aspect of the curriculum?
 - What misunderstandings might these errors represent?
 - How might the structure of the test question have shaped the response?

Using a combination of the NAPLAN data, school data and professional judgment, teachers may then test these hypotheses to see whether they are valid or whether there is more to be thought about and investigated. Teachers can then plan lessons related to the general areas where students seem to need help. Teachers can also make judgments about teaching approaches and curriculum.

The professional conversations that are part of this process are the most effective and powerful way to use the data, as they are the vehicle for developing shared understandings.

Placing the tests in the assessment context

The results from the NAPLAN tests should be seen as only one input into a school's assessment program. Various forms of assessment are needed to inform the different stages of the teaching and learning cycle. Principals and teachers should keep in mind that NAPLAN is a point-in-time, timed test that can only cover a few curriculum features.

The results from a school's own assessments of students should be consistent with the NAPLAN test results. If the test results are different from what was expected, consider possible reasons. The results of the tests may indicate aspects of student performance that need further investigation within the classroom, using other forms of assessment.

An item with a low facility rate (percentage correct) may not necessarily indicate a problem in teaching and learning. It may be that this was simply a difficult item for all students in this cohort across Australia.

Other NAPLAN reports

In addition to the State reports, the following reports are produced about the performance of Queensland students who sit the NAPLAN paper tests:

SunLANDA Online

Since 2015, student data has been released on the QCAA School Portal using the SunLANDA Online interface. Access to SunLANDA as application software is also still available on the QCAA website.

SunLANDA Online provides class and school information in an electronic form that permits customised spreadsheet generation by users. In addition, it shows representative samples of students' incorrect responses to constructed responses where applicable. Hyperlinks from within SunLANDA Online lead to the QCAA's test item analysis. Information on how to use this service is available at: www.qcaa.qld.edu.au/p-10/naplan/test-reporting-analysis/sunlanda/accessing-navigating-sunlanda.

Test item analysis

These PDF documents contain an analysis of each test item. They can be downloaded directly from the QCAA website: <https://www.qcaa.qld.edu.au/p-10/naplan/test-reporting-analysis/test-item-analysis>. A school Brief Identification Code (BIC) and password is required to access these documents. The analysis reproduces each item followed by expert analyses of how the item operated. It shows the distractors presented in multiple-choice items and explains students' reasoning.

School and class reports

The NAPLAN school and class reports are supplied electronically on the secure section of the QCAA website. These reports are accessible only with the school's Brief Identification Code (BIC) login and password. Individual student reports are distributed to schools as printed copies.

School reports

The QCAA issues NAPLAN school reports giving information about each school's performance. They provide a summary of year-level performance as well as performance by gender, language background and Indigenous status in the following fields:

- distribution of scaled scores
- distribution of achievement bands
- school and state means
- participation of the group.

The school report positions a school's performance within the state on a graph that is shaded to show the range of performance for the middle 60% of Queensland students together with the state mean.

Class reports

The QCAA issues NAPLAN class reports that show the performance of every student on every item. Under the name of each student is recorded the items they had correct and incorrect. They also show students' responses to constructed-response items.

The class report also gives the:

- percentage correct for each item for the class and state, and by gender

- scaled scores for each student
- performance bands for each student.

Individual student reports

The QCAA issues individual student reports to schools after the tests. Schools receive one printed report for each student to distribute to parents/carers.

ACARA reports

As well as the Queensland reports from the QCAA, national reports are available from the website of the Australian Curriculum Assessment and Reporting Authority (ACARA). The *NAPLAN National Summary Report* and the *NAPLAN National Report* allow states and territories to place the achievement of their students in relation to their peers across the nation. This is system-level information and is publicly available.

Literacy

Writing

Stimulus (writing prompt) Years 7 & 9

YEAR 7 AND YEAR 9

New technology

Write a persuasive text about a piece of technology that has been or will soon be invented that will make life so much better.

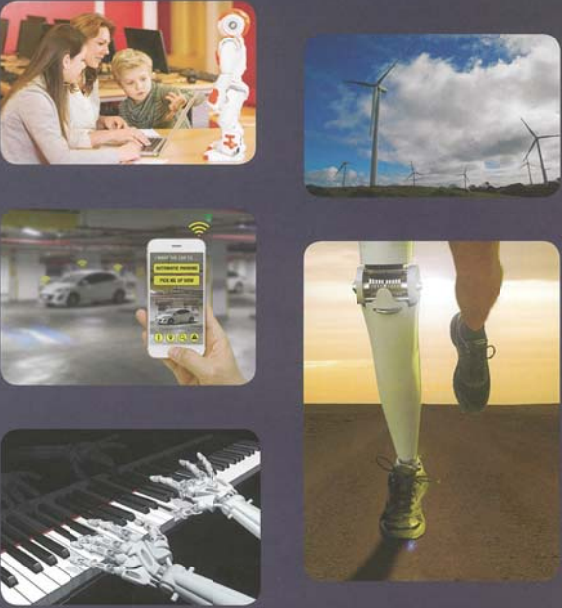
Persuade a reader why this device or machine, such as a self-driving car, a drone helicopter or a fingerprint lock for a bike will benefit people.

You can use your own idea or you can use an idea on this page.

- **Start with an introduction.**
An introduction lets a reader know what you are going to write about.
- **Write your opinions on the topic.**
Give reasons for your opinions.
Explain your reasons.
- **Finish with a conclusion.**
A conclusion sums up your reasons so that a reader is convinced of your opinions.

Remember to:

- plan your writing
- use paragraphs to organise your ideas
- write in sentences
- choose your words carefully to convince a reader of your opinions
- pay attention to your spelling and punctuation
- check and edit your writing.



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About the task

In 2018, the NAPLAN Writing test was based on the persuasive genre. As has been the case since 2015, two prompts were used: one for Years 3 & 5 and another for Years 7 & 9. The test conditions and administration remained the same as in previous years, i.e. teachers delivered the same spoken instructions and read the text aloud to students. Working independently, students had to plan, compose and edit a written response. Students were allowed five minutes to plan, thirty minutes to write their script, and a further five minutes to edit and complete the task. Three pages were provided for students to write a response.

The 2018 prompt for Years 7 & 9 was titled *New technology*. Students were asked, in the textual component of the prompt, to:

- Write a persuasive text about a piece of technology that has been or will soon be invented that will make life so much better.
- Persuade a reader why this device or machine, such as a self-driving car, a drone helicopter or a fingerprint lock for a bike will benefit people.

Students were told they could use their own idea, or an idea mentioned or pictured on the page.

Additional textual information was provided. This named the structural components, and further defined these elements, e.g. *Start with an introduction. An introduction lets the reader know what you are going to write about.* Other notes were also provided in relation to the conventions associated with the writing task, e.g. write in sentences, check and edit your writing. Five images were shown on the prompt sheet, covering a range of possible topics/ideas such as wind turbines (technology for renewable energy), mechanical artificial limbs that use technology and an app for a mobile phone.

The prompt was relatively open-ended, allowing students to base their writing on one or more of the images provided, or compose their text around their own idea.

Markers for this Writing test were trained using the national persuasive writing marker training package, delivered as part of ACARA's national assessment program. Markers were recruited and trained in accordance with national protocols. Registered Queensland teachers marked the NAPLAN Writing test scripts. All markers applied the 10 criteria and related standards from the marking rubric. Writing test scripts were marked on screen in all states and territories. Stringent quality-control measures were applied to the marking of student scripts, including a prescribed percentage of scripts to be double-marked, and the daily application nationally of control scripts for all markers. As part of the Queensland marking operation for 2018, referee marking continued, further ensuring marking reliability. There was also provision for appeal over individual Writing test scores after the results were released.

An earlier version of the NAPLAN Persuasive writing marking guide is available at: www.nap.edu.au/_resources/Amended_2013_Persuasive_Writing_Marking_Guide_-With_cover.pdf

Performance

The 2018 writing prompt was particularly suited to many students in Years 7 and 9 as technology is something they know about, have an interest in and have seen develop throughout their life. For many students it would be something they have looked at in their STEM subjects and studies. The prompt did provide some examples of technology for students and did also allow students to select a topic of personal interest with which they had some familiarity. It appears that few students had difficulty in finding a subject on which their text could be based, and the majority of students went with suggestions from the prompt, particularly self-driving cars and fingerprint locks for bikes.

Students typically introduced their subject (e.g. a piece of technology that will make life better) in an opening paragraph, stating broadly why this piece of technology would make life better, then provided information regarding the uses and benefits of the technology. The danger here was that following the introduction, some students tended to move into informative text rather than persuasive text. Conclusions then tended to focus on simple re-statements of main points referred to in the body of the text. Often there was a very close parallel between the wording of the introduction and conclusion.

If students did follow the informative line, without taking a stance on the subject in question, they were deemed to be 'off genre', an outcome that may have had considerable impact on their scores. Some students, particularly Year 9 students, adopted a 'review'-type response, tending to provide information rather than persuasive argument, defending their choice of technology. Typically, this information was sandwiched between an introduction and conclusion which reflected some persuasive elements. These students were deemed to be 'on-genre', though their final scores were impacted by the absence of persuasive elements throughout their texts.

In most cases, students in Year 9 showed competence with the persuasive form. Introductions provided more natural orientation for the reader, with quite passionate statements of position often presented.

For example:

Have you ever looked up into the sky and wondered, how did we get here? Have you ever seen the utter smile on a child's face when they gaze at the effervescent stars? The love of the outside universe makes our world seem so wonderful. Imagine if we had a vehicle that could take you there. Ten billion light years in a flash! We could travel anywhere, see anything, and explore to our absolute limits. Would you like to spend every day like a holiday? Zooming through the stars, fascinating your inner child? Well, in the future, you're in for a must-have treat! (Year 9 student)

As the Writing test is an 'on-demand' task, where students are given 5 minutes planning time, 30 minutes to write their persuasive task and 5 minutes for editing and completing their work, students frequently lacked some detail about the technology they were referring to. Though time in a demand writing task obviously affects the way in which student writing concludes, many responses included only a brief summation or re-statement of the text. Those students who, in conclusion, challenged the reader to fully consider the arguments presented, produced highly effective closure to their writing, as the following extract demonstrates:

The importance of self-driving cars is undeniable. However, sometimes they can do more harm than good. Self-driving cars would make the world a much safer place and ensure that cars are used more for good than for bad. By assisting tired drivers, transporting unwell passengers and preventing reckless driving, these cars will make the world a better place. The world is always changing, sometimes for better, sometimes for worse, and self-driving cars are one significant invention that could improve the amazing world that we call home. (Year 9 student)

Though scored independently, the strong connection between the criteria of *audience*, *persuasive devices* and *ideas* in this type of task rewards students who are prepared to take on 'the big idea' and explore this in a more engaging way. Alternatively, students who wrote with passion and commitment about a subject they knew very well, produced credible responses.

The NAPLAN marking rubric also allocates significant score points to the skills areas of *sentence structure*, *punctuation* and *spelling*. Persuasive writing almost encourages the use of adverbial clauses and phrases indicating causation and condition. Stronger scripts showed variety and control over complex sentence forms, with 'punchy' simple sentences (and even sentence fragments) occasionally used for marked effect. Punctuation, particularly sentence boundary punctuation, still indicates room for improvement. The use of 'run-on' sentence forms, often associated with the use of 'splice commas' as breaks, is a significant factor. Much of this has to do with the shift from oral to written language modes. The formalities associated with writing, in test conditions and elsewhere, need constant attention in writing programs. Punctuation can easily become a casualty of contemporary communication forms if it is not dealt with through a rigorous and contextualised writing program, closely associated with reading good-quality published texts.

In 2018, the use of improbable data and quotations was less evident than in previous years. Teachers should be wary of approaches to writing that suggest the inclusion of implausible data or dubious supporting evidence attributed to some 'invented' authority. Markers are trained to accept at face value what is on the page, but irrelevant or erroneous information does little to support a writer's point of view, particularly at the higher end of the writing spectrum.

Undoubtedly this year, students were more acquainted with the topics of their choice and could 'flesh out' arguments with plausible supporting detail. Some students' field knowledge led them to write quite detailed texts designed to persuade readers to agree with the stance they were taking regarding the type of technological improvements that could be made.

Virtual Reality becoming a Reality

Technology has become an essential part of daily life for billions world-wide. In the last few decades, countless life-changing machines and technological advances have made the impossible possible. One such invention is the rapidly growing medium of virtual reality. It is a unique platform that is perfect for a wide variety of tasks. These range ^{from} teaching environments to creative outlets. Virtual reality is incredibly versatile in what it can do, making it one of the most important technological creations of recent times. With consistent improvements in quality, there is little virtual reality cannot do.

Virtual reality presents a powerful and intriguing teaching device. Currently, all that's needed ^{to access it} is a headset or a monitor, so it could easily be integrated into the classroom. It is the perfect learning tool for visual and interactive learners. Programmers can create any environment within virtual reality. They could display the inside of a human heart, enabling students to get a true sense of how blood and oxygen are pumped through their bodies. Alternatively, they could show two animals running ^{beside each other} and enable students to see themselves trying to out run a cheetah. There are many other ways virtual reality can be utilised, and its capabilities will only expand with more time. It is valuable and exciting in its youth, and will quickly become an important part of any teaching program.

While virtual reality teaching is still in its very early stages, there is another side to VR. That's far more popular. It is the artistic side of the medium, especially gaming, painting and sculpting. Virtual reality gaming is an incredibly powerful tool. Using a simple headset and controls, players can 'become a character' in a more real sense than ever before. While books and movies can get part of the way, V.R. is able to fully immerse players into a whole new world. They can be the hero, fighting ³⁰ dragons, or they can fly to a never-before-seen planet. These aspects of virtual reality are unique and ground-breaking, and more elements are constantly presented.

Virtual reality is still being developed, but it is easy to see why it must become an essential part of everyday life. It has many uses, and will have even more in the future, ^{currently} including a massive range of futuristic teaching capabilities and a one-of-a-kind artistic ³⁰ platform. VR technology will only get better from here. It is a technology that everyone must keep an eye on from this point onwards. Virtual reality is quickly becoming a reality.

Commentary on sample script

This persuasive text is focused on the development of virtual reality. It is authoritative and impassioned. The writer adopts an enthusiastic, expert stance, setting the scene by first mentioning technology and technological advances, leading into the main subject of the script, *the rapidly growing medium of virtual reality*.

The script orients the reader and provides sufficient information for the reader to follow easily. The introductory paragraph informs the reader that this *unique platform ... is perfect for a wide variety of tasks*, and then proceeds to mention some of the ways virtual reality can be used. These tasks are then elaborated in the body of the text.

The language choices throughout the text are chosen to encourage the reader to continue reading to understand fully what the writer is informing them of. The writer takes the stance of an expert and enthusiastically imparts the knowledge she has of the uses of virtual reality.

Following the introduction, the writer describes the use of virtual reality as a *powerful and intriguing teaching device* and provides examples of different ways it could be integrated into the classroom. This paragraph is developed to show the reader that virtual reality is not a toy but rather a valuable teaching tool as it can clearly show students a range of realistic environments as they are learning about various topics.

The next paragraph informs the reader that *virtual reality teaching is still in its very early stages* and that *there is another side to virtual reality that's far more popular*. The writer then informs the reader of the *artistic side of the medium, especially gaming, painting and sculpting*.

The tone of the text is lively and excited as the reader is informed of all the possibilities for virtual reality. Throughout the text the writer's position is clear. The student effectively uses some words and phrases to make the point that virtual reality is continually developing and persuades the reader that it has many extensive and varied attributes that are continually developing.

Structurally, all textual elements work in a coordinated and deliberate fashion as the argument is built. The text is highly cohesive, and the choice of text connectives and conjunctions strongly contribute to this, together with lexical chains that effectively link the stages of the text. The vocabulary is well suited to the overall flavour of this text.

Similar to other aspects of the text, the skills areas of spelling, punctuation, paragraphing and sentence structure are well constructed, making this script an excellent example of a demand writing task.

Language conventions

Spelling

Results and item descriptions

The percentage columns give the facility rate (percentage correct).

Item	Answer	QLD %	Aust. %	Description
Proofreading — Error identified				
1	garage	90.71	91.57	Corrects the spelling of a two-syllable word ending in -age.
2	research	89.20	88.78	Corrects the spelling of a two-syllable word with ear.
3	subscription	75.48	74.88	Corrects the spelling of a three-syllable word with scrip.
4	familiar	61.44	62.68	Corrects the spelling of a three-syllable word ending in -liar.
5	incredible	68.70	68.66	Corrects the spelling of a multisyllable word ending in -ible.
6	cereal	52.71	54.93	Corrects the spelling of a three-syllable homophone starting with ce-.
7	distressed	47.99	49.56	Corrects the spelling of a two-syllable word ending in -ssed.
8	monitored	42.19	43.79	Correctly spells a 3-syllable word ending in -ored.
9	feud	35.53	38.94	Corrects the spelling of a single-syllable word with eu.
10	maintenance	26.05	26.63	Corrects the spelling of a three-syllable word ending in -enance.
11	adjoining	25.94	26.70	Corrects the spelling of a three-syllable word starting with adj-.
12	grievances	28.50	28.01	Corrects the spelling of a three-syllable word ending in -ances.
Proofreading — Error not identified				
13	telescope	80.00	80.78	Corrects the spelling of a three-syllable word starting with tele-.
14	wisdom	71.77	75.05	Identifies and corrects an error in a two-syllable word with is.
15	specialist	66.77	67.15	Identifies and corrects an error in a three-syllable word ending in the suffix -ist.
16	infectious	57.63	59.87	Identifies and corrects an error in a three-syllable word ending in the suffix -ious.

17	casualty	54.67	55.71	Identifies and corrects an error in a three-syllable word ending in -ualty.
18	suburban	46.29	47.70	Identifies and corrects an error in a three-syllable word ending in -an.
19	scenario	35.95	36.56	Identifies and corrects an error in a multisyllable word starting with sc-.
20	juvenile	33.64	33.96	Identifies and corrects an error in a three-syllable word ending in -enile.
21	dissatisfaction	23.10	23.05	Identifies and corrects an error in a multisyllable word with the prefix dis-.
22	intrepid	7.87	8.70	Identifies and corrects an error in a three-syllable word ending in -id.
23	guarantee	19.55	20.67	Identifies and corrects an error in a three-syllable word starting with gu-.
24	indispensable	20.82	21.11	Identifies and corrects an error in a multisyllable word ending in -able.
25	plagiarism	2.59	3.57	Identifies and corrects an error in a multisyllable word with iar.

About the test

The 2018 Year 9 test focused on the following spelling features:

- affixes: *familiar*, ***incredible***, *monitored*, ***maintenance***, *grievances*, ***adjoining***, *specialist*, ***infectious***, ***casualty***, *suburban*, ***dissatisfaction***, *intrepid*, ***indispensable***
- affixes with consonant alternation: *subscription*, *wisdom*
- complex vowels and consonants: *research*, ***feud***, ***guarantee***, *plagiarism*
- Greek and Latin elements: ***telescope***, ***scenario***, ***juvenile***
- open/closed syllable: ***garage***
- homophone: *cereal* (*serial*)
- tense inflection: ***distressed***.

Both sets of questions in the Spelling test use proofreading formats. The target words in the first set are misspelt and identified by being circled. Those in the second set are misspelt but not identified, so students need to find them inside sentences containing other words (distractors) that are correctly spelt. These supplied misspellings may lead students to spell differently from when they write to dictation or compose their own sentences.

Performance

Compared to the national average, Year 9 students in Queensland performed only slightly lower on most words. They did marginally better on the words *research*, *subscription*, *incredible* and *grievances*. Compared to previous Queensland cohorts, this year's Year 9 students performed slightly better in Spelling.

Between 10% and 15% of students omitted any response to the last eight words. The reading load and vocabulary demand in these items was high.

Facility rates fell below 10% for the words *plagiarism* and *intrepid*, indicating that few students in the cohort have knowledge of the words.

Common student errors show the following:

- Instead of treating suffixes as meaning chunks with set spellings, many students still attempt to 'sound them out'. This resulted in misspellings such as *specialest*, *monitered*, *protestor* and *familier*.
- Many students still struggle with the suffixes *-ible/-able* and *-ence/-ance* (e.g. *incredible*, *indispensable*, *maintenance*, *grievances*).
- There was a surprising difficulty with the word *infectious* (*infectous* 11%, *infectus* 3%).
- Many students misspelt the base word parts of the words *grievance*, *casualty* and *scenario* (*grieve*, *casual* and *scene*).
- For some of the items in the second set (e.g. *guarantee*, *dissatisfaction* and *specialist*), a very large percentage of students changed the correct spelling of the distractor words (e.g. *refrigerator*, *protesters* and *referred*) instead of trying to spell the target word.
- Many students may have been influenced by the supplied misspelling of the word *distressed* as *distrest*.

Implications for teaching

Test-wiseness

Ignore supplied errors: Students should avoid being influenced by the supplied misspelling of a word. Instead, they should focus on their own spelling knowledge and strategies.

Word study

Students need to build a mature vocabulary. Word study belongs in all subjects, not just English. Students should learn:

- the pronunciation of advanced words and their look on the page
- the meaning of their component morphemes (word parts)
- other words built from the same stem.

Students may learn to avoid inappropriate 'sounding out' strategies if they know that suffixes and inflections have grammatical effects and stable spellings. For example, they should be able to reason that the final syllable of *specialist* contains the agent noun suffix *-ist*, not the superlative adjective suffix *-est*.

In the light of the poor facility with the word *infectious*, teachers in subject English might focus on the regular patterns for suffixes. For example, the suffix *-ous* belongs with other suffixes that form adjectives, such as *-y*, *-able*, *-acious*, *-ic*, *-al*, *-ish*, and *-est*. From another angle, *-ous* words can be divided into those with a /sh/ sound, e.g. *suspicious*, and those without, e.g. *indigenous*. Exceptions, rarities and oddballs should be noted as well as the general patterns.

Proofreading

Proofreading skills should be taught as an authentic writing skill. This will incidentally help students read test questions carefully and avoid being misled by supplied errors.

QCAA resources

Full analysis of student performance and error patterns for each item is published in the SunLANDA program: <https://www.qcaa.qld.edu.au/p-10/naplan/test-reporting-analysis/sunlanda/accessing-navigating-sunlanda> and as PDF documents:

<https://www.qcaa.qld.edu.au/p-10/naplan/test-reporting-analysis/test-item-analysis>. A school BIC and password is required to access each year level document.

Grammar and punctuation

Results and item descriptions

The percentage columns give the facility rate (percentage correct).

Item	Answer	QLD %	Aust. %	Description
26	B	92.22	93.05	Identifies subordinating conjunction to complete a complex sentence.
27	D	83.96	85.15	Identifies a sentence from a review containing an opinion.
28	C	82.59	81.44	Identifies a simple sentence that needs quotation marks.
29	C	83.84	84.68	Identifies a complete sentence.
30	B	79.31	79.48	Identifies the reference for a pronoun in a short passage.
31	A	69.58	70.51	Identifies a sentence with an embedded clause error.
32	B	58.78	55.76	Identifies the correct punctuation of quoted speech with internal attribution.
33	D	64.59	64.29	Identifies the sentence expressing certainty.
34	A	59.44	60.47	Identifies the correct use of a colon.
35	A	71.39	74.23	Identifies the correct placement of commas in a list with single items.
36	D	61.82	63.01	Identifies a sentence that correctly combines information from three separate sentences.
37	A	53.82	53.45	Identifies a preposition in a simple sentence.
38	C	37.37	38.18	Identifies a sentence without a parallel construction error.
39	B	40.71	40.32	Identifies an adverb in a simple sentence.
40	B	35.59	35.42	Identifies the correct use of apostrophes of possession for plural nouns.
41	D	47.98	46.62	Identifies an adverb in a complex sentence.
42	D	20.50	20.16	Identifies a colon as the correct punctuation required in a complex sentence.
43	B	25.20	23.36	Identifies the correct use of dashes for a list in a simple sentence.
44	C	24.84	23.12	Identifies an embedded adjectival clause.
45	A	24.41	25.43	Identifies jargon in a compound sentence.
46	A	27.73	29.66	Identifies that a verb is missing in a simple sentence.
47	BD	11.84	11.47	Identifies semicolons as correct punctuation to

				separate extended items in a list.
48	C	19.03	18.52	Identifies the non-finite verb to complete a complex sentence.
49*	2,1,2,2	15.95	15.52	Identifies four sentences as being in active or passive voice.
50	D	13.61	10.47	Identifies a noun phrase in a complex sentence.

* Item 49: Answers in sequence: passive, active, passive, passive.

About the test

The NAPLAN Language conventions items test sentence-level, clause-level and word-level skills. The test does not cover the curriculum. Instead, it tells how a large number of students perform on a small range of tasks. Standardised tests can, however, suggest broad trends across a cohort. At the level of individual students, NAPLAN results can supplement classroom assessments and guide teachers to important points of grammar and punctuation that need revisiting. The Language conventions test comprises 50 items in total: 25 Spelling items and 25 Grammar and punctuation items.

In 2018, about one half of the questions were focused on students' knowledge of the terminology of grammar and punctuation terms (the 'metalinguage'). The number and complexity of metalinguage questions has been increasing over the years.

Students needed to know the terms for the parts of speech to answer the following questions:

- Item 37, Item 39, Item 41 and Item 46 on adverbs
- Item 50 on noun groups.

These questions sometimes also referred in the distractor options to the names of other parts of speech, e.g. preposition, conjunction, adjective, verb, noun.

Students needed to know the terms for structural parts of sentences to answer the following:

- Item 31 on clause order (i.e. a relative clause follows the subject noun)
- Item 44 on adjectival (relative) clauses
- Item 49 on active/passive voice.

These questions sometimes also referred in the distractor options to the names of other structural elements, e.g. adverbial clause, main clause, adverbial group and adjectival phrase

Students needed to know the terms for punctuation marks to answer the following questions:

- Item 28 on quotation marks for direct speech and attribution
- Item 42, Item 34 and Item 47 on colons and on semicolons (in complex lists)
- Item 43 on dashes (i.e. em rules separating a parenthetical phrase).

The following questions could be answered **without** knowledge of metalinguage:

- Item 26 and Item 36 on conjunctions
- Item 29 on subordinate clauses (as fragments needing a main clause)
- Item 30 on pronoun reference
- Item 32 on punctuating and attributing direct speech
- Item 35 on list commas

- Item 38 on parallel structure
- Item 40 on punctuating possessive pronouns, plural possessives and contracting 'it is'
- Item 46 on the subject–verb–predicate structure and on gerund phrases
- Item 48 on participial clauses.

For information about the full range of grammar knowledge that Year 9 students should have, refer to the Australian Curriculum English.

Performance

Year 9 students in Queensland performed marginally higher than the national average in many items. Compared to previous Queensland cohorts, Year 9 students performed a little better.

Many students, including many of the high-performing ones, struggled with the metalanguage questions. The reasons for this vary. In the case of Item 42, the terms 'colon' and 'semicolon' do not themselves create the low facility but the lack of the important knowledge of how to use these signs. However, the low facility for Item 49 is likely due to the special grammar analysis task it sets. In the case of Item 50, even many students who understood the terms 'noun group' and 'main clause' did not know how to interpret this example. The low facility and the weak statistical 'fit' for Item 46 (on parts of speech) may also have been due to the item format. Few students omitted responses to any items, except for Item 45.

Implications for teaching

Grammar and punctuation is not a separate area of reading and writing, but an essential component. The contexts of language in use are pivotal to a full understanding of how syntax, grammatical and punctuation conventions operate. Wherever possible, meaning and purpose in texts should be emphasised, noting how particular language conventions contribute to making meaning.

The Australian Curriculum English allows for the progressive teaching knowledge and understanding of grammatical concepts and punctuation conventions. Teaching the curriculum implies teaching the terminology of grammar (metalanguage) where this helps the effective use of language to communicate in different contexts and for different purposes.

QCAA resources

Full analysis of student performance and error patterns for each item is published in the SunLANDA program: <https://www.qcaa.qld.edu.au/p-10/naplan/test-reporting-analysis/sunlanda/accessing-navigating-sunlanda> and as PDF documents: <https://www.qcaa.qld.edu.au/p-10/naplan/test-reporting-analysis/test-item-analysis>. A school BIC and password is required to access each year level document.

Reading

Results and item descriptions

The percentage columns give the facility rate (percentage correct).

Item	Answer	QLD %	Aust. %	Description
<i>Dear Mr Mitchell</i>				
1	C	95.53	95.97	Identifies the author of a persuasive letter.
2	C	85.69	88.98	Interprets a cohesive reference used in a persuasive letter.
3	B	96.79	97.33	Locates directly stated information in a persuasive letter.
4	B	16.15	16.88	Evaluates the purpose for the use of italics in a persuasive letter.
5	D	89.47	89.93	Interprets techniques used by an author to position a reader in a persuasive letter.
6	A	27.95	29.49	Interprets a rhetorical device in a persuasive letter.
<i>Our world of weird and wonderful plants</i>				
7	D	79.85	78.01	Identifies the reader's expected reaction in an information text.
8	A	91.56	92.02	Identifies the way the writer introduces an information text.
9	C	64.24	65.56	Identifies the writer's focus in a section of an information text.
10	B	75.34	76.34	Identifies information not given in an information text.
11	D	86.16	87.30	Identifies an exception from comparisons in an information text.
12	B	43.56	45.01	Identifies a definition in an information text.
<i>Graffiti – is it art?</i>				
13	A	88.32	89.57	Identifies the reasoning behind an author's point of view in an argument.
14	D	53.27	57.38	Identifies a persuasive technique in an argument.
15	B	63.19	65.22	Identifies evidence used to support a conclusion in an argument.
16	C	24.20	26.46	Identifies an example of emotive language in an argument.
17	B	67.92	70.24	Identifies a belief in an argument.

18	A	59.33	60.77	Compares attitudes represented in two arguments.
<i>The high life</i>				
19	D	67.95	69.81	Identifies the reason for information in brackets in a factual description.
20	C	40.19	39.94	Interprets directly stated information in a factual description.
21	C	71.67	73.98	Interprets information in a factual description.
22	B	39.88	40.14	Identifies the effect of an intensifier in a factual description.
23	B	62.26	63.90	Interprets information in a factual description.
24	C	53.37	56.49	Interprets information in a factual description.
<i>The forging</i>				
25	C	84.11	85.9	Interprets the purpose of text structures and language features in a narrative.
26	D	45.92	45.30	Interprets figurative language in a narrative.
27	B	52.31	53.78	Interprets technical language in a narrative.
28	D	69.64	69.92	Identifies a character's mood change across a narrative.
29	A	60.19	61.04	Synthesises information from dialogue and description in a narrative.
30	B	47.60	48.23	Interprets dialogue in the final two paragraphs of a narrative.
31	4,1,3,2	29.42	29.31	Identifies the sequence of events in a narrative.
<i>Computer gaming: sport or not?</i>				
32	B	42.97	45.12	Evaluates the central argument in a discussion transcript.
33	C	67.28	69.22	Identifies supporting evidence in a discussion transcript
34	D	72.59	74.57	Identifies a speaker's argument in a discussion transcript.
35	A	48.10	50.41	Analyses how a speaker uses persuasion in a discussion transcript.
36	A	52.20	53.28	Identifies how a speaker uses persuasion in a discussion transcript.
37	AC	6.88	7.40	Synthesises information from across a text in a discussion transcript.
<i>Lost for words</i>				
38	AE	24.08	26.17	Identifies how a title reflects aspects of a speech with internal commentary.

39	BD	40.96	44.10	Infers a character's perspective in a speech with internal commentary.
40	C	74.50	76.87	Analyses the effect of language devices in a speech with internal commentary.
41	D	39.67	39.91	Interprets an allusion in the middle of a speech with internal commentary.
42	BCD	6.4	7.31	Identifies a character's perspective in a speech with internal commentary.
43	ABGH	29.69	32.22	Interprets an implied meaning in a speech with internal commentary.
<i>Nonsense rules!</i>				
44	A	47.35	48.31	Analyses the effect of how the writer introduces a topic in a complex essay.
45	C	56.28	58.76	Extracts the meaning of less familiar vocabulary in the middle of a complex essay.
46	B	41.16	42.35	Extracts key information and ideas from the middle of a complex essay.
47	C	27.93	28.33	Identifies the writer's opinion of the subject of a complex essay.
48	AD	12.5	13	Applies understanding of complicated ideas from a complex essay.
49	T, F, T, T	25.58	26.66	Identifies the validity of key information from across a complex essay.
50	CDE	4.2	4.46	Infers multiple meanings from the title of a complex essay.

About the test

The 2018 Reading test consisted of 50 items based on eight reading magazine units spanning several genres. There were no short-response items. Most items were in a standard four-option, multiple-choice format. However, this year's test included eight items (up from four last year) which required multiple responses. Sometimes the student is instructed to *Choose two*, sometimes the student is instructed to *Choose all*, which further increases difficulty for students. It is important to note that this type of item does not reward partial accuracy. As in previous years, there was also an item that required students to indicate how information is sequenced in a text.

As is usually the case, there was a pattern of increasing difficulty as students progressed through the test. Facility rates decline as students progress from the first reading unit to the last as the reading demands generally increase, distractors appear to be more sophisticated and item construction becomes more complex.

Rather than categorise the reading units by genre, it is more useful to categorise the items by question type (see table below). This is partly because many of the reading stimulus texts contain features of multiple genres. Furthermore, the error analysis of individual items sometimes shows that students will select a distractor because it alludes to a genre. Even capable students will be drawn to these distractors.

The Year 9 Reading test contained two link units with Year 7: *The forging* and *Lost for words*.

Categorising items by *question type* puts the focus on the more relevant reading strategies. For example, an examination of the question types makes it apparent that simply underlining or highlighting text may only be useful for one of the seven question types — Literal-recall questions. See the *Implications for teaching* section in this text, and SunLANDA for a range of reading strategies that are specific to the types of questions.

	Question type:	The reader is asked to:
Literal	Recall	Recognise or recall information.
	Translation	Change information into a different form — it might involve paraphrasing the ideas or restating them in terms or forms other than those in the text.
Text-based inferential	Interpretation	Identify the relationships among ideas, definitions, facts and values — these would involve such relationships as comparisons and cause and effect; they involve a minimum of higher-order thinking as the reader/learner needs only to respond to and manipulate ideas in the text.
Higher-order (Context-based) inferential	Application	Solve real-life problems by extrapolating what is in the text — readers/learners need to combine ideas from the text with prior knowledge.
	Logical analysis	Analyse and judge the quality of the logic inherent in the text — readers/learners might, for example, identify fallacies or points of view represented in a text.
Creative	Synthesis	Respond to a problem or idea with original and creative thinking.
	Evaluation	Make judgments with respect to specific criteria.

Performance

The level of difficulty increased as students progressed through the Reading test, with the first units having slightly higher facility rates than the following units. This was more pronounced in other year levels than the Year 9 Reading test. Performance in link items from *The forging* and *Lost for words* ranged from 0.4% to 13.12% higher when compared to Year 7 performance.

The first two units, *Dear Mr Mitchell* and *Our world of weird and wonderful plants*, contained a range of question types and were generally quite easy for many Year 9 students except for Item 4. Item 4 required students to evaluate the purpose of the use of italics which required a context-based inference. Therefore, part of the difficulty is that it requires higher-order inference. Another reason this was quite difficult for students is that one of the distractors was very difficult to eliminate. Even though this item appeared to refer to one very specific part of the text, any context-based inferential question requires an understanding of the whole text. This is a point which many of the students appeared to miss.

The third unit, *Graffiti – is it art?*, was more difficult than the first two units. This unit had two items that required creative thinking (question type). One of these, Item 16, proved to be the most challenging in this unit. Several variables contributed to the difficulty of this item. Options were drawn from both texts, *Graffiti is art* and *Graffiti is vandalism*. Students were required to know what is meant by ‘emotive language’ and how to apply it in the context of the text. The item analysis shows that able students struggled with this item. They had difficulty with the fact that to identify language as emotive requires reading that language and its context. Many students attempted to apply this in a lexical sense without referring to the context. Once again, the implication is that students should respond to Reading test questions using the text.

The *High life* and *The forging* had comparable average facility rates and question types. *The forging* was a link unit from Year 7. On average, Year 9 students performed 7% better than Year 7 students. The lowest item facility rate in these two units was Item 31, which interestingly was a literal-translation question type. This is likely because it is the first non-conventional item format in the Reading test. It requires students to sequence four events. Many students were tempted to sequence the events as they occurred in the text. These students did not understand that the first event was one that appeared in the second paragraph of the text. This is often the case with sequencing items in the Reading test.

The next unit, *Computer gaming: sport or not?*, was slightly more difficult for students. It contained the third most difficult item in the Year 9 Reading test. Item 37 had a 6.88% facility rate because it required higher-order inference and because it required students to select two keys. Higher-order inference questions are generally more challenging because students are required to synthesise information from across a text.

The next unit was *Lost for words*. Year 9 students appeared to find Item 42 very challenging as they barely achieved a higher facility rate than Year 7 students for the same item. There were several difficulties with this item. One was that the item required the selection of an unspecified number of correct options, three in this instance. Another is that readers had to identify multiple perspectives of the writer in a text where some readers struggled to distinguish the writer's actual perspective from what she stated. The very low facility rate for this item suggests that many Year 9 students may not have developed their ability to identify a character's perspective and/or manage item types like this.

In the final unit, *Nonsense rules!*, Item 50 was the most difficult item in the entire test (facility rate 4.2%). The omission rate was barely higher by the end of the test, so most students were still engaged throughout the test. The complexity for students is increased through the requirement to supply an unknown number of responses. In this case, three of the five options constitute the key. Option E, which was selected by the fewest students, may have been missed because students found it difficult to combine ideas from the entire text. Also, as has been seen with other items, many students attempt to respond to an item using prior knowledge without referring to the text adequately.

In summary, Year 9 students had 50 questions in the Reading test: twelve of these were literal, twenty-two required text-based inference, fourteen required context-based inference and two were creative. There appeared to be a slight increase in the average facility rate for creative type questions, but it should be noted there were only two of this type of question. Otherwise it can be seen that item difficulty increases with the degree of inference required. The item analysis shows that complexity also depends on the reading difficulty of the stimulus text and the sophistication of the distractors.

Question type	Average facility rate	Number of items
Literal (recall and translation)	72.10	12
Text-based inference (interpretation)	52.64	22
Context-based inference (application and logical analysis)	39.01	14
Creative	41.77	2

The items that involved purpose, tone and character generally had lower facility rates than literal and lower-order inferential items. This is because they required higher-order reasoning and comprehension. Students need to form an understanding of the whole text as well as pay attention to subtle clues in the text that help them make the inferences. As a result, implications for teaching should reflect these demands. Year 9 students did not perform as well in all the Year 7 link items as might be expected.

Implications for teaching

The lower facility rates on non-literal items demonstrates the importance of giving students strategies to help them make inferences as they read, i.e. to make statements about the unknown based on the known.

The challenge for teachers is to get students to annotate texts in the classroom and discuss them in groups so that they can see how all the parts of the text contribute towards the meaning of the whole. This is the time to discuss the patterns in the text (e.g. cause and effect), identify connections between ideas in the text, the two or three main parts of the text and how the parts contribute to the overall meaning. All of this should occur before they begin a close study of the text. Students will handle the distractors in the items much better if they are clear about the subject matter and the purpose of the text before they proceed to the items.

Teachers can encourage students to read for pleasure and recreation to extend their knowledge of themselves and the world around them. Reading develops empathy for characters and people in difficult situations. Students also need to be able to participate confidently in a close study of a text, to check for fallacies and persuasive techniques, and to identify emotive language and literary techniques. The goal should be for students to be discerning and capable readers and confident speakers and writers about those texts.

The complexity of the reading process is made visible when students discuss texts and how they arrive at their personal understanding of the text. Teachers are the facilitators of this process, not the leaders. Their focus should be on:

- finding authentic texts which engage their students
- providing a range of genres and texts from classic or traditional texts to texts with postmodern elements
- promoting higher-order questioning of texts (both set texts for special study and unseen texts for close study)
- reading aloud to students to promote reading for pleasure (sometimes at Year 9 this is forgotten)
- developing an awareness of how the parts of the text combine to create a whole through both semantic (links between the ideas) and syntactic (grammatical links) cohesion
- encouraging students to make inferences as they read (an informed guess backed by evidence or a statement about the unknown based on the known)
- encouraging the link between reading and writing by asking students to regularly write analytical paragraphs about an aspect of what they have read (which includes a controlling central idea) in response to the question, e.g. Can this character be trusted? Is there a shift in tone in this text?
- encouraging students to look deeper into a text by drawing on analytical skills, e.g. explore gaps and silences, consider writer bias, look at contradictions within the text, look for themes, hidden purposes and so on and re-read the text from another perspective (e.g. a contemporary, feminist, eco-critical perspective).

QCAA resources

Full analysis of student performance and error patterns for each item is published in the SunLANDA program: <https://www.qcaa.qld.edu.au/p-10/naplan/test-reporting-analysis/sunlanda/accessing-navigating-sunlanda> and as PDF documents: <https://www.qcaa.qld.edu.au/p-10/naplan/test-reporting-analysis/test-item-analysis>. A school BIC and password is required to access each year level document.

Additionally, further advice and support can be found in: QCAA 2015, *Beyond NAPLAN: How to read challenging texts*, Beyond NAPLAN series: www.qcaa.qld.edu.au/downloads/p_10/naplan_read_challenging_texts.pdf.

Numeracy

Results and item descriptions

The Numeracy strands from the Australian Mathematics Curriculum are abbreviated as follows: Number and Algebra (NA); Measurement and Geometry (MG); Statistics and Probability (SP).

All items are worth one score point. However, there is a range of difficulty across the items.

The percentage columns give facility rates (percentage correct) for each item.

Item	Strand	Answer	QLD %	Aust. %	Description
Calculator-allowed items					
1	MG	A	93.05	92.93	Identifies the top view of a collection of cylinders.
2	MG	A	81.59	82.48	Converts measurements to a common unit to compare lengths.
3	SP	B	77.23	78.13	Describes the effect of removing an outlier on the mean of a set of data.
4	SP	B	65.73	68.61	Selects the most likely results in a spinner experiment.
5	NA	50	72.01	72.52	Calculates the difference between two integers.
6	NA	A	70.16	72.93	Calculates the profit made in a simple financial plan.
7	MG	B	62.78	65.39	Solves a timetable problem involving 24-hour time.
8	NA	88.5	30.54	32.07	Divides and multiplies a decimal by whole numbers in context.
9	NA	A	78.98	79.04	Selects a graph showing a constant rate of decrease to match a real situation.
10	MG	B	77.41	79.22	Calculates the perimeter of a kite from one side length and the ratio of its sides.
11	NA	A	60.29	62.78	Uses multiples of whole numbers to solve a problem in context.
12	MG	E	58.43	59.48	Locates the relative position of a point on a grid given compass directions.
13	NA	D	52.16	55.02	Solves a division problem to determine the remainder.
14	NA	C	61.35	64.11	Subtracts decimals and multiplies by one million.
15	NA	C	71.18	71.94	Evaluates an algebraic expression by substituting a value into the equation.
16	MG	D	60.36	59.96	Identifies the face opposite a given face on the net of a cube.

17	MG	C	58.61	59.97	Calculates the size of angles using properties of angles on parallel lines.
18	SP	C	66.10	67.51	Identifies the most representative sampling strategy.
19	SP	B	47.89	49.36	Calculates the probability of the complement of an event using a Venn diagram.
20	NA	B	50.98	51.33	Adds and subtracts fractions with unlike denominators in context.
21	NA	C	39.73	41.94	Solves a problem using simple ratios and the addition and subtraction of decimals.
22	NA	B	37.74	37.76	Determines the rule for a linear relationship on the Cartesian plane.
23	NA	D	40.99	43.24	Selects the appropriate order of operations to give a correct expression.
24	SP	150	21.55	25.01	Identifies the median from a stem-and-leaf plot.
25	NA	D	34.19	36.39	Uses algebraic reasoning to solve a problem involving the four operations.
26	MG	95.8	23.01	27.28	Calculates the circumference of a circle given its diameter.
27	NA	C	34.64	35.90	Compares items of different size by determining the unit price of each.
28	MG	D	24.79	27.14	Identifies the triangle congruent to a given triangle and correctly labelled.
29	SP	D	44.40	48.62	Uses data from a sample to predict the most likely results for a population.
30	MG	693	6.62	9.06	Calculates the volume of a solid given the area of its faces.
31	MG	C	18.98	18.76	Calculates the capacity of a cylinder to solve a problem involving rates.
32	MG	AC	19.26	19.38	Rotates and reflects a kite on the Cartesian plane and identifies its properties.
33	MG	42	12.39	14.05	Calculates the length of a diagonal of a rectangle using multiplicative reasoning.
34	NA	25	8.87	9.58	Interprets a non-linear graph to solve a multistep problem.
35	SP	0.16	7.71	8.31	Calculates the probability of a two-step experiment using a tree diagram.
36	NA	CF	9.09	12.25	Applies the index laws to a numerical expression with integer indices.
37	SP	0.15	4.87	5.96	Uses complementary events and the sum of probabilities to solve a problem.
38	MG	144	6.77	8.29	Solves a problem using the perimeter of a

					trapezium and parallel lines properties.
39	NA	4	6.03	7.22	Calculates the simple interest rate given the principal, time and amount accrued.
40	NA	112.5	4.99	6.47	Uses direct proportion to solve a problem involving profit.
Non-calculator items					
1	NA	C	76.37	78.61	Divides a decimal by a whole number.
2	NA	D	59.65	62.05	Converts between fractions and percentages.
3	NA	530	27.70	31.27	Calculates a simple fraction of a quantity where the result is a whole number.
4	NA	C	34.52	36.27	Solves a problem involving the combination of multiples of 3, 7 and 9.
5	NA	36.55	20.66	23.23	Multiplies, adds and subtracts decimals without digital technologies.
6	NA	3 213 000	15.26	16.92	Calculates the cost of an item after a percentage discount.
7	NA	CE	17.41	19.53	Applies the laws and properties of arithmetic to algebraic terms and expressions.
8	NA	1.6	8.19	10.25	Solves a problem involving dividing a quantity in a given ratio.

About the test

The Year 9 Numeracy test consisted of 48 items covering concepts and skills from the Australian Mathematics Curriculum across two tests — a Calculator-allowed (CA) paper with 40 items and a Non-calculator (NC) paper with 8 items. However, not all items in the Calculator-allowed paper required the use of a calculator.

Student results for Numeracy in Year 9 are reported as a single score.

The distribution of the 48 items across the Australian Mathematics Curriculum strands was:

- Number and Algebra (26 items)
- Measurement and Geometry (14 items)
- Statistics and Probability (8 items).

All 8 items in the Non-calculator paper were from the Number and Algebra strand.

Over the entire test of 48 items, 32 (67%) were multiple-choice, with the remaining 16 items (33%) requiring students to arrive at their own answers (constructed responses). Interpretation of tables, graphs or diagrams was required in 27 of the 48 items (56%). The other 21 items (44%) were word problems that usually incorporated numerals within the information and question stems.

Performance

This year, 95.4% of Queensland Year 9 students scored at or above the national minimum standard (Band 6), compared to 95.5% of Australian students overall. Queensland scale score shows a steady trend for improvement over the past decade:

The percentage of students who correctly answered Calculator-allowed items ranged from 93% (Item 1) down to 5% (Items 37 and 40). Item 37 required students to use knowledge of the properties of complementary events and subtract their probabilities to deduce the probability of a stated event. The other most difficult question for students, Item 40, involved extrapolating data in a table involving two separate rates, one decreasing and one increasing, to determine the profit made on selling several items.

For the Non-calculator items, the percentage of students answering items correctly ranged from 76% (Item 1) down to 8% (Item 8). Item 8 required students to determine the number of grams of copper, to be expressed as a decimal, which is found in 8 grams of an alloy. The composition of the alloy was presented as a three-part ratio involving three metals.

Twenty-one of the 40 Calculator-allowed items were answered correctly by 50% or more of Queensland students. Similarly, four out of the 8 Non-calculator items were answered correctly by 50% or more of the Queensland students.

There was a difference of more than 4% between the facility rates (the percentage of correct responses) of the national cohort and the Queensland cohort of students on only two Items (26 and 29). In Item 26, Queensland students performed below the national cohort by 4.3% and in Item 29 Queensland students were 4.2% lower than the national mean. Item 26 was from the Measurement and Geometry strand. Students had to calculate the circumference of a drum, given its diameter in centimetres, to one decimal place. Item 29 was from the Statistics and Probability strand. Students needed to take a proportion from a sample of students and apply this to the whole population.

Queensland students performed equal to or above the national cohort on four items of the Calculator-allowed paper: Items 1, 16, 22 and 31. In the Non-calculator paper, Queensland students performed below the national cohort on all eight items. However, the difference in facility rates was small (less than 4%).

While the majority of students attempted to answer all questions, many students omitted the more challenging items towards the end of each paper. The items with more than a 25% omission rate were items 33, 35, 37, 38, 39, and 40 in the Calculator-allowed paper and Item 8 in the Non-calculator paper. All of these items required a constructed response rather than selecting an answer from given multiple-choice options. These items are designed to differentiate student performance — to provide opportunities for higher performing students to demonstrate their ability to reason mathematically. Item 35 had the highest omission rate of 35%. It involved a tree-diagram with four branches, the probabilities on each branch were stated as fractions with different denominators and the answer was to be expressed as a decimal to two decimal places. This item required students to interrogate several pieces of written information regarding a swimming team with two coaches, the number of swimmers they each train and the random selection of one swimmer into a swimming team.

Implications for teaching

Over the entire Numeracy test of 48 items, there were only four Items (22, 28, 32 and 36 of the Calculator-allowed paper) that could be considered as purely mathematical. The other items presented problems set in real-life contexts. Therefore, it is important that teachers provide learning activities and tasks that, where possible, allow students to investigate mathematics and develop solutions to problems that are set in real-life contexts. It is also recommended that

teachers seek to combine mathematics learning from multiple strands. This will enable students to make connections between different areas of mathematics.

Non-calculator items

In each of the 8 items in the Non-calculator paper, Queensland students performed below the national cohort of Australian students. This suggests that working through mathematical problems without using calculators is an area where Queensland students could improve.

There are times when the calculations required in a problem are too complex or the values involved have many decimal places. In these instances, a calculator may be essential. However, there are many situations where a calculator is not needed. For example, in straightforward number problems and patterns, in some problems involving fractions and percentages, and in money problems where change or discounts have to be calculated.

Where appropriate, teachers should encourage their students to perform the calculations required in solving mathematics problems without using calculators, at least initially. Students could then be encouraged to use calculators as a means of checking their own manually obtained answers. This practice would also be beneficial for students as a life-skill, as a means of estimating answers or quantities and for those times when a calculator is not available. Another benefit from manually calculating answers would be to help students decide on the reasonableness of answers that are obtained when they do use a calculator.

Tables, graphs, diagrams

In the 2018 Year 9 Numeracy test, most of the items in the Calculator-allowed paper (56%), required the interpretation of tables, graphs, diagrams or other images. Students' visual literacy influences their ability to make sense of mathematical information presented in different representations.

It is important that students be given frequent opportunities to interpret and construct a variety of graphical displays including pie graphs, column graphs and line graphs.

Students also need to be able to interpret various data displays, such as stem-and-leaf plots, data tables and Venn diagrams. It is also recommended that students become familiar with using tree diagrams and interpreting timetables.

Many of the diagrams in NAPLAN tests are geometric in nature. They often involve nets, perspectives, lengths, perimeters, areas, volumes, grids, scales, angles, locations and directions. Students need exposure to learning experiences that involve a wide variety of geometric representations.

The ability to recognise and interpret the different ways that data and information can be represented is an important element of numeracy. Teachers can lead class discussions about the visual representations of mathematical concepts and provide opportunities to engage with a wide variety of mathematical diagrams. This will help students develop the skills and experience required to interpret them. Even questions without visual stimulus may be conceptualised by sketches, diagrams or models.

Word problems

In the 2018 Year 9 Numeracy test, the Non-calculator items were mostly word problems while 44% of the Calculator-allowed items were word problems; overall, this is a significant proportion of the test. Therefore, it is important that students have frequent exposure to mathematics problems presented in sentence form, interspersed with numbers. Students need to practise reading written text in numeracy contexts, interpreting the information and deciding what is required to be done.

Many students find word problems challenging as they require reading, interpreting and deciding what procedures to use. Students need exposure to a wide range of word problems that involve different combinations of mathematical operations. They need to become familiar with the language associated with operations and the terminology that is used when dealing with word problems. Money-related problems, for example, lend themselves to written descriptions. Consequently, students need to have a good grasp of the language of money, e.g. terms such as discount, profit, loss, sale price, percentage increase and decrease.

When dealing with a word problem, students should be encouraged to read the whole question more than once. The first time to get a general idea of what it is about, with subsequent readings to identify important information and what the question is asking to be done. Circling, underlining or highlighting key information may help to identify important details for students. Sorting written information into a more useful form such as drawing a sketch or diagram, a table or a list can be a valuable habit for students to adopt. The ability to translate word problems into arithmetical or algebraic expressions or equations is also an important element in solving word problems.

Problem solving

Students need frequent exposure to challenging problems to develop the confidence and skills to find the solutions. Teachers should consider teaching and learning through presenting students with problems that range from simple to complex, familiar to unfamiliar and single-step to multistep. Problem solving should be incorporated into learning activities, not treated as a separate or add-on activity. It should be included in a range of learning experiences, both in real-life and purely mathematical contexts. Where possible, teachers are encouraged to pose student maths problems that are derived from different curriculum areas.

It is possible to equip students with approaches to apply when a numeracy problem is encountered. This will give them confidence to attempt and develop solutions. Students can be equipped with problem-solving approaches where they feel enabled to:

- Identify the problem (What am I being asked to do?)
- Analyse the problem (What do I have to work with?)
- Take appropriate action (Select and apply procedures to solve the problem)
- Reflect on the answer (Check my answer — Does it work? Is my answer reasonable? Are there other correct answers?).

While all steps are important, 'Does my solution work?' and 'Is my answer reasonable?' are particularly beneficial in test situations, especially when students are asked to construct a response.

Students should be habitually asked to consider the reasonableness of their answers. See for example Calculator-allowed Items 8, 26, 33 and 38 where the constructed responses could readily be checked for reasonableness. Knowing a problem-solving strategy will help when students are required to provide constructed responses. The items with the highest omission rates in the 2018 Numeracy test were those involving constructed responses.

Test-wiseness

Teachers should instruct students to answer all multiple-choice items. Even if this involves guessing some answers before the test time runs out, it is better practice to answer these items than not answering at all. An educated guess may sometimes be correct.

As well as the multiple-choice questions, students should expect constructed-response items to be included in the test. They need to have the confidence to commit to providing their own numerical answers. Students can develop their mathematical confidence by regularly engaging

with a wide variety of tasks that require them to produce numerical responses, either independently or in groups.

The low facilities and high omission rate for word problems can be improved. The literacy features that are typical of numeracy contexts should not be unfamiliar or daunting to students. Literacy teaching should not be confined to the English class.

Difficult items

Teachers are encouraged to closely inspect the items that Queensland students found the most difficult. Generally, this would be the items where the Queensland facility rates were below 30%. Teachers should consider whether it was the content covered and/or the style of these questions that made them difficult and what the implications are for their students.

QCAA resources

Full analysis of student performance and error patterns for each item is published in the SunLANDA program: <https://www.qcaa.qld.edu.au/p-10/naplan/test-reporting-analysis/sunlanda/accessing-navigating-sunlanda> and as PDF documents: <https://www.qcaa.qld.edu.au/p-10/naplan/test-reporting-analysis/test-item-analysis>.

A useful reference for the teaching of spatial reasoning and geometric properties is given here: QSA, 2005, *Mathematics: About space*, https://www.qcaa.qld.edu.au/downloads/p_10/kla_maths_info_space.pdf